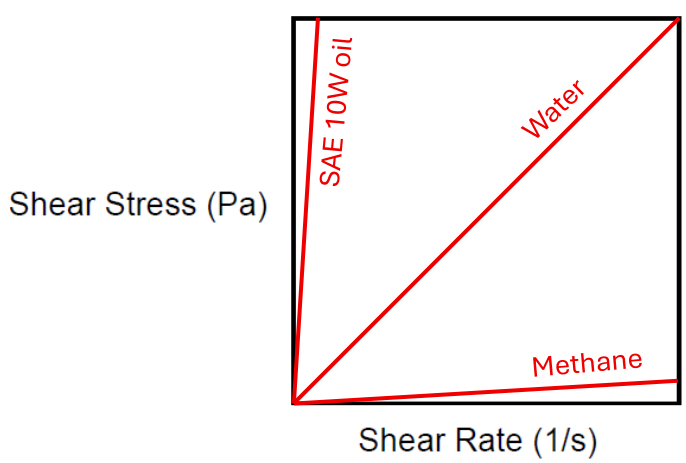
CHEMICAL ENGINEERING 2O04
Midterm #1

Duration 120 minutes (7:00pm – 9:00pm) D.R. Latulippe
McMaster University Term Test Thursday February 12th, 2026

QUESTION 1 (6 MARKS)

Answer the following short questions in the space provided:

a.	Show that the equation below for a venturi meter (from slide 40 of PDF_03) is dimensionally consistent: **(2 marks)** $$Q = \frac{\pi}{4} D_2^2 \sqrt{\frac{2 \cdot (P_1 - P_2)}{\rho \left(1 - \frac{D_2^4}{D_1^4}\right)}}$$	$$Right\ hand\ side\ [=] m^2 \sqrt{\frac{Pa}{kg/m^3}}$$ $$[=] m^2 \sqrt{\frac{kg/ms^2}{kg/m^3}} [=] m^2 \sqrt{\frac{m^2}{s^2}} [=] m^2 \frac{m}{s} [=] \frac{m^3}{s}$$
b.	Sketch (no calculations needed) the three shear stress versus shear rate profiles (at ambient temperature) for these Newtonian fluids: SAE 10W oil Water Methane **(2 marks)**	According to figure on slide 23 of PDF_00 - Viscosity SAE 10W oil = 0.1 Pa · s - Viscosity Water = 0.001 Pa · s - Viscosity Methane = 0.00001 Pa · s Newtonian fluids have linear relationship between shear stress & shear rate with proportionality constant (aka slope) being the viscosity 

c.	Describe the difference between a Venturi meter and a rotameter in terms of their ability to be used for automated control in any piping system. **(1 mark)**	Only the venturi meter can be used for automated control – the rotameter requires an operator to go and look at the position of the internal float within the vertical tube.
d.	Describe the difference between a Venturi meter and a rotameter in terms of their orientation (i.e. horizontal, vertical) when used in any piping system. **(1 mark)**	The rotameter must be oriented vertically (since it works on balance of drag, gravity, & buoyancy forces) – the venturi meter is normally oriented horizontally, although it can be oriented vertically for certain applications.

QUESTION 2 – 5 MARKS

Chinese hamster ovary (CHO) cells are a family of immortalized cell lines that are widely used in suspension bioreactors to produce therapeutic proteins. A colleague asks you to help with designing a gravity settler device to remove the CHO cells from the product stream that is withdrawn from the bioreactor – the key parameter they need is the terminal settling velocity in the protein-rich product stream.

a. Using the information below (for process operating at ambient temperature), calculate the terminal settling velocity (mm/s) of a perfectly spherical CHO cell.
(4 marks) *HINT: Since CHO cells are quite small, they will settle very slowly!*

Cell volume = 1.15e-15 m^3

Cell density = 1.09 g/cm^3

Product stream density = 1.03 g/cm^3

Product stream viscosity = 0.0018 Pa*s

Rearrange the equation from slide 16 of PDF_03 to solve for terminal velocity:

$$V_T = \frac{2{R_S}^2(\rho_s - \rho_f)g}{9\mu}$$

Where $\rho_s = 1.09\, {}^{g}/_{cm^3} = 1090\, {}^{kg}/_{m^3}$

$$\rho_f = 1.03\, {}^{g}/_{cm^3} = 1030\, {}^{kg}/_{m^3}$$

Calculate R_s from the given volume $\forall_s = \frac{4}{3}\pi {R_S}^3$

$$R_s = \sqrt[3]{\frac{3}{4\pi}\forall_s} = 6.5 \times 10^{-6} m$$

And so

$$V_T = \frac{2(6.5 \times 10^{-6})^2(1090 - 1030)9.81}{9(0.0018)} = 3.1 \times 10^{-6}\, {}^{m}/_{s}$$

$$\boldsymbol{V_T = 3.1 \times 10^{-3}\, {}^{mm}/_{s}}$$

b. Circle the correct statement:
(1 mark)

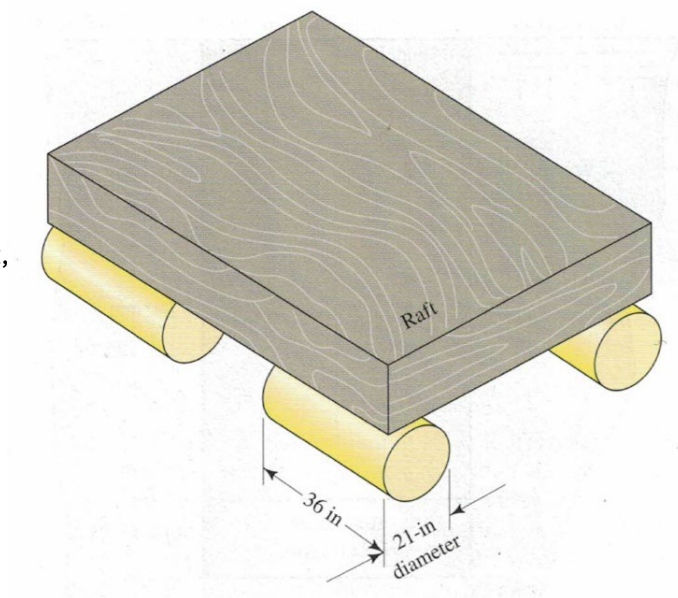
CHO cells will settle faster at a lower temperature

CHO cells will settle slower at a lower temperature

At lower temperature, product stream viscosity will increase and thus settling velocity will decrease.

QUESTION 3 – 5 MARKS

The accompanying figure shows a raft made of a wooden platform attached to four hollow plastic drums (one in each corner).



a. If each drum weighs 13.5 kg, what is the maximum mass (kg) that the raft can support, so the platform and anything placed on it, at the point when the drums are completely submerged in fresh water (at 15 °C) but the platform is still above the water.
(4 marks)

A force balance on the raft in its final position yields the following equation:

$$F_{buoyancy} = F_{gravity}$$

Where the buoyancy term is mostly based on the water displaced by the four plastic drums and so must calculate the volume of each plastic drum:

$$\forall_{drum} = \frac{\pi}{4} D^2 L = \frac{\pi}{4}\left(21in\frac{0.0254m}{1in}\right)^2\left(36in\frac{0.0254m}{1in}\right) = 0.204m^3$$

Now back to the force balance $\rho_{water} g(4\forall_{drum}) = (4m_{drum} + m_{platform} + m_{added})g$

or

$$m_{platform} + m_{added} = \rho_{water}(4\forall_{drum}) - (4m_{drum})$$

From slide 20 of PDF_00 $\rho_{water} = 999\ {}^{kg}/_{m^3}$

$$m_{platform} + m_{added} = 999(4(0.204m^3)) - (4(13.5kg))$$

$$\boldsymbol{m_{platform} + m_{added} = 761kg}$$

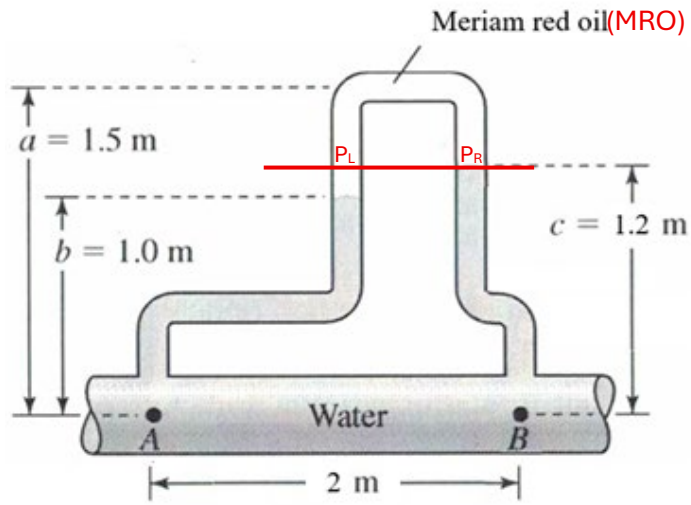
b. Would your answer from part a) increase or decrease if the raft was instead placed in the ocean? No calculations needed.
(1 mark)

It would increase - since seawater has higher density (1025 kg/m^3) than fresh water.

QUESTION 4 – 5 MARKS

As shown in the accompanying figure, ambient temperature (i.e. 20 °C) water is flowing through a horizontal pipe to which an inverted manometer (containing Meriam red oil) is attached.

a. Calculate the pressure difference (Pa) between points A and B.
(4 marks)



At location shown on figure: $P_L = P_R$

Expression to relate P_L and P_A: $P_A = P_L + \rho_{MRO} g(c-b) + \rho_{water} g(b)$

Expression to relate P_R and P_B: $P_B = P_R + \rho_{water} g(c)$

Combining these three equations:

$$P_A - \rho_{MRO} g(c-b) - \rho_{water} g(b) = P_B - \rho_{water} g(c)$$

Rearrange to solve for $P_B - P_A$:

$$P_B - P_A = \rho_{water} g(c) - \rho_{MRO} g(c-b) - \rho_{water} g(b)$$

$$P_B - P_A = \rho_{water} g(c-b) - \rho_{MRO} g(c-b)$$

$$P_B - P_A = (\rho_{water} - \rho_{MRO}) g(c-b)$$

From slide 30 of PDF_03 $\rho_{MRO} = 827 \; {}^{kg}/_{m^3}$

From slide 20 of PDF_00 $\rho_{water} = 998 \; {}^{kg}/_{m^3}$

$$P_B - P_A = (998 - 827)9.81(0.2)$$

$$\boldsymbol{P_B - P_A = 335 \; Pa}$$

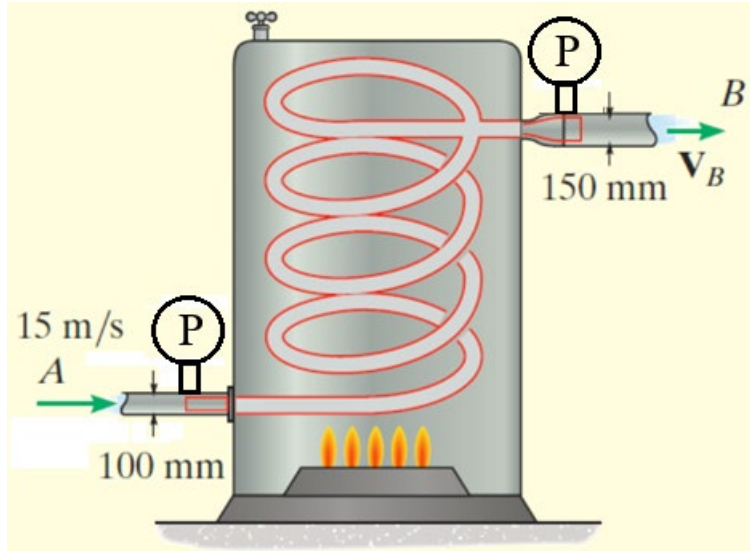
b. Based on answer in part a, circle the correct statement:
(1 mark)

Water is flowing from point A towards point B

Water is flowing from point B towards point A

Since $P_B > P_A$

QUESTION 5 – 7 MARKS



a. Starting from the general form of the mass balance equation for ANY control volume with multiple mass flow rate inlets and outlets (given below), derive the appropriate equation for steady-state flow of air within the internal piping of the gas heater that is shown in the accompanying figure.
(2 marks)

$$\sum(\dot{m}_{inlet}) - \sum(\dot{m}_{outlet}) = \frac{dm}{dt}$$

There is a single inlet, a single outlet, and the system is at steady state, so the general mass balance simplifies to:

$$\dot{m}_{inlet} - \dot{m}_{outlet} = 0$$

or

$$\dot{m}_{inlet} = \dot{m}_{outlet}$$

b. Determine the average velocity (m/s) of the heated air stream that exits at position B given the information below:

At position A: air temperature = 20 °C; air pressure = 17.2 psi-g; average velocity = 15 m/s

At position B: air temperature = 75 °C; air pressure = 7.0 psi-g

(5 marks)

Express each mass flow rate term as product of density, average velocity, and cross-sectional area

$$\rho_A V_A A_A = \rho_B V_B A_B$$

Rearrange to solve for V_B and plug in formula for cross-sectional area of pipe

$$V_B = V_A \frac{\rho_A}{\rho_B} \frac{\frac{\pi}{4} {D_A}^2}{\frac{\pi}{4} {D_B}^2} = V_A \frac{\rho_A}{\rho_B} \frac{{D_A}^2}{{D_B}^2}$$

From Slide 39 of PDF_02: $\rho_{gas} = \frac{P \cdot M_W}{R_U \cdot T}$ $(where\ R_U = 8.314 \frac{m^3 \cdot Pa}{mol \cdot K})$

And so $$V_B = V_A \frac{\frac{P_A \cdot M_W}{R_U \cdot T_A}}{\frac{P_B \cdot M_W}{R_U \cdot T_B}} \frac{{D_A}^2}{{D_B}^2}$$

or

$$V_B = V_A \frac{P_A}{P_B} \frac{T_B}{T_A} \frac{{D_A}^2}{{D_B}^2}$$

Remember for ideal gas law that must be absolute pressures & temperatures, so

P_A = 219, 914 Pa

P_B = 149,588 Pa

T_A = 293.15 K

T_B = 348.15 K

and thus $$\boldsymbol{V_B = 11.6\ {}^{m}/_{s}}$$

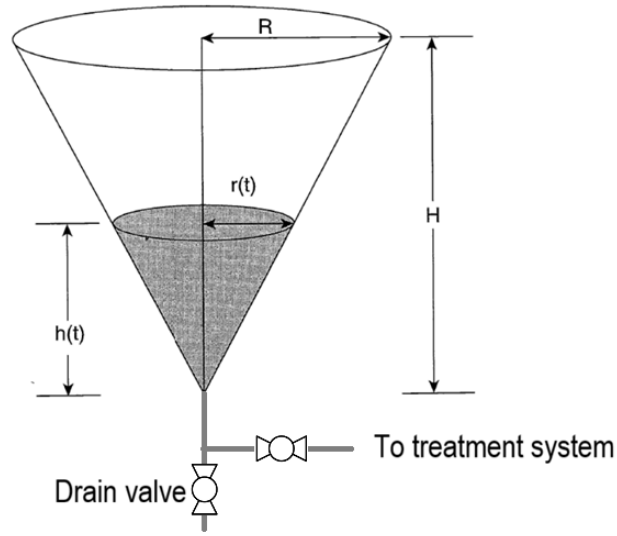
QUESTION 6 – 11 MARKS

As mentioned in lecture, an inverted conical tank (with height H and radius R) with a bottom drain valve (with internal diameter d) is quite useful for lab-scale water treatment systems because the tank can be easily emptied by simply opening the drain valve. For a fully-opened drain valve, the average velocity of water leaving the tank can be approximated as $\sqrt{2gh_{tank}}$

where

g is the gravitational constant (9.81 m/s^2)

h_{tank} is the height of water in the tank (m)



a. Starting from the general mass balance equation for ANY control volume with multiple mass flow rate inlets and outlets (given below), derive the appropriate equation for the change in liquid level (i.e. dh_{tank}/dt) when the drain valve is in the fully-opened position.

NOTE: For this scenario, the other valve in the figure above is closed so that none of the water is being sent to the treatment system.

$$\sum(\dot{m}_{inlet}) - \sum(\dot{m}_{outlet}) = \frac{dm}{dt}$$

(3 marks)

There are no inlets and only one outlet, so the general mass balance simplifies to:

$$-\dot{m}_{outlet} = \frac{dm}{dt}$$

According to the content on slide 11 of PDF_04, since the conical tank has variable cross - sectional area, the mass balance becomes:

$$-\rho Q_{outlet} = \rho \frac{d\forall}{dh}\frac{dh}{dt}$$

The same conical tank appeared in Question 4 of Assignment 2, which the solutions show that:

$$\frac{d\forall}{dh} = \pi \frac{R^2}{H^2} h^2$$

And so the mass balance becomes:

$$-Q_{outlet} = \pi \frac{R^2}{H^2} h^2 \frac{dh}{dt}$$

b. Solve the differential equation from part a) to get the correct expression for the height of water (i.e. $h_{tank}(t)$) while the conical tank is being drained from some initial height (h_o).
(5 marks)

Recognizing that the outlet flow rate equals the average velocity (expression given above) times the cross-sectional area of the valve, the mass balance now becomes:

$$-\sqrt{2gh}A_{valve} = \pi \frac{R^2}{H^2} h^2 \frac{dh}{dt}$$

By separation of variables:

$$-\sqrt{2g}\frac{H^2}{\pi R^2} A_{valve} dt = h^{3/2} dh$$

Then plugging in limits of integration

$$-\sqrt{2g}\frac{H^2}{\pi R^2} A_{valve} \int_0^t dt = \int_{h_o}^{h} h^{3/2} dh$$

Then integrating both sides

$$-\sqrt{2g}\frac{H^2}{\pi R^2} A_{valve} t = \frac{2}{5}\left[h^{5/2} - {h_o}^{5/2}\right]$$

And rearranging to solve for h

$${h_o}^{5/2} - \frac{5}{2}\sqrt{2g}\frac{H^2}{\pi R^2} A_{valve} t = h^{5/2}$$

$$\boldsymbol{h = \left[{h_o}^{5/2} - \frac{5}{2}\sqrt{2g}\frac{H^2}{\pi R^2} A_{valve} t\right]^{2/5}}$$

THIS IS NOT REQUIRED – can check the validity of above equation by plugging in t = 0:

$$h = \left[{h_o}^{5/2} - 0\right]^{2/5}$$

which simplifies to

$$h = h_o$$

c. Consider the following scenario – after the tank has been completely emptied, it is to be refilled with a different source water for another treatment study. However, the drain valve is accidentally left open while the new source water is being constantly added (via some flexible tubing looped over the top rim of the conical tank) at a rate of 30 L/min. Calculate the maximum height that the water will reach inside the conical tank given the following:

Tank height (H) = 30 cm

Tank radius (R) = 12 cm

Inner diameter of bottom drain valve (d) = 2 cm

(3 marks)

The maximum height occurs at steady-state condition, which is mathematically:

$$\dot{m}_{inlet} - \dot{m}_{outlet} = 0$$

or

$$\dot{m}_{inlet} = \dot{m}_{outlet}$$

$$\rho Q_{inlet} = \rho Q_{outlet}$$

$$Q_{inlet} = Q_{outlet}$$

$$Q_{inlet} = \sqrt{2gh_{max}} A_{valve}$$

$$Q_{inlet} = \sqrt{2gh_{max}} \frac{\pi}{4} d^2$$

or

$$h_{max} = \left(\frac{Q_{inlet}}{\sqrt{2g} \frac{\pi}{4} d^2} \right)^2$$

Now plug in the values:

$$h_{max} = \left(\frac{30 \frac{L}{min} \left(\frac{1m^3}{1000L} \right) \left(\frac{1min}{60s} \right)}{\sqrt{2 \cdot 9.81 \, {}^{m}/_{s^2}} \frac{\pi}{4} (0.02m)^2} \right)^2$$

$$\boldsymbol{h_{max} = 0.129 \, m}$$

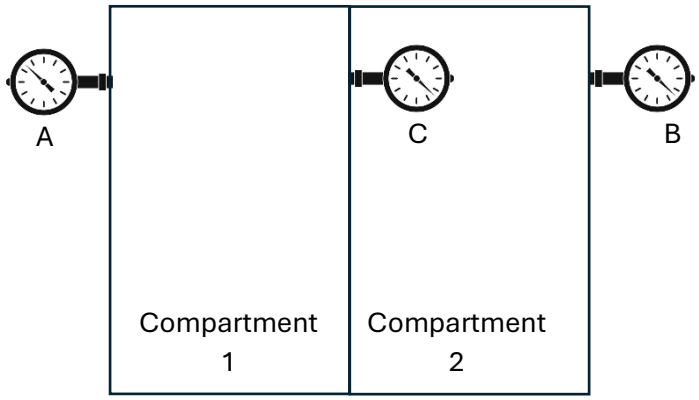
So what is happening in the system?

- If the height of water is less than 0.129 m, it will be filling faster than it is draining
- If the height of water is greater than 0.129m, it will be draining faster than it is filling
- If the height of water is exactly 0.129m, it is filling and draining at the same rate

QUESTION 7 – 3 MARKS

As shown in the accompanying figure, a closed vessel is divided into two equal compartments by an internal wall – there are three identical compound pressure gauges (labeled A, B, and C) with dials that display both positive and negative pressures. If gauge A displays +207 kPa and gauge B displays -39 kPa, what pressure (kPa) will be displayed on gauge C (connected to compartment 1, but located inside compartment 2).

NOTE: Assume standard atmospheric pressure.



From reading on gauge A, the absolute pressure inside compartment 1 is 306 kPa (i.e. 101 + 207).

From reading on gauge B, the absolute pressure inside compartment 2 is 62 kPa (i.e. 101 – 39).

Gauge C will display the pressure in compartment 1 relative to the pressure in compartment 2, which is mathematically 306 – 62 = 246 kPa.